Executive Summary

Vicon Ground-Truth Validation of the AutoPos UWB Self-Localisation System

Erlangen, 28 May 2026 | Zekai Xiao

All "§" and "Table" references link to the full report `main_EN.pdf`; numbers are shared with it.

This report presents the first external ground-truth validation of the AutoPos UWB anchor self-localisation system against a Vicon motion-capture reference (28 May 2026, MaD Lab, FAU Erlangen-Nürnberg; §1). Eight DWM1001C-Dev anchors run broadcast SS-TWR with UWB-only inter-anchor self-calibration; a 12-camera Vicon system (sub-millimetre, 0.18 mm median world error) provides truth. The evaluation covers 24 static tag positions, 17 dynamic RotoArm captures, and an offline UWB+IMU fusion replay. All claims are dataset-bound to this measurement.

Key results at a glance

Quantity	Result	Full report
Static 3D accuracy — median / P95 / RMSE	72.7 / 171.5 / 109.8 mm	§5.1, Table 5
Static horizontal / vertical median	37.4 / 61.9 mm	Table 5
Internal repeatability σ_{3D} — unfiltered → filtered	67.1 → 18.6 mm	Table 6, §6
Anchor self-cal. residual / apparent scale	92.8 mm / +4.36%	§3, §4.2
Dynamic RotoArm floor — track P50, persists	105.8 mm	§7, Table 9
Best UWB+IMU fusion — track P95	231.8 → 112.1 mm	§11, Table 18

Headline static accuracy. Under anchor-locked rigid Vicon registration, the production v4-io/T4/F0 pipeline achieves **72.7** mm **median 3D error** over 24 static positions (full breakdown in the table above; §5.1, Table 5). This is sub-decimetre median static localisation from a range-only self-calibrated array in this geometry — not a universal UWB specification, since layout, tag placement, vertical aperture, and propagation all affect it.

Anchor self-calibration and the apparent scale bias. The v4-io layout reproduces the Vicon anchor geometry to 92.8 mm median residual (§3, Table 2). The dominant layout-error mode is not random shape distortion but a coherent +4.36% **apparent scale** (mean inter-anchor excess 120.5 mm/pair). A per-anchor decomposition (§4.2, Table 4) traces this to common-mode per-device range bias (mean 94.6 mm/device, all eight terms positive) that the bounded production delay terms under-absorb, so the residual appears geometrically as layout expansion. A re-parameterised solver with a free common mode recovers metric scale to ≈1% from UWB ranges alone, leaving sub-percent scale for bench delay calibration.

Key scientific finding — delay–layout coupling. A four-way ablation (layout source × residual-correction treatment; §4, Table 3) shows that optically surveyed Vicon anchor coordinates *degrade* tag localisation unless the residual delay/range-bias corrections are re-estimated in the Vicon coordinate frame (77.7 mm vs. 108.9 mm RMSE in the ablation estimator). The fitted corrections are **frame-conditioned** and cannot be transplanted across coordinate frames. To our knowledge this is the first explicit four-way ablation showing that externally measured ground-truth coordinates cannot simply replace a self-calibrated UWB layout — the most transferable result of the campaign and the intended core of a methods publication.

Vertical error is systematic, not geometric. The static error is vertically dominated (61.9 mm vertical vs. 37.4 mm horizontal median). A range-only DOP analysis rules out geometric observability: VDOP ≈ HDOP over the measured volume (§12, Table 21), and repeatability is *not* vertically dominated whereas accuracy is, so the deficit is systematic bias, not amplified noise. A height-driver audit identifies the uncorrected **tag-side delay** as the dominant height-dependent lever; the expanded anchor layout partially *masks* it. The 72.7 mm headline therefore benefits in part from this cancellation: holding tag delay at zero on the clean metric-corrected layout gives 109.5 mm median, while an oracle same-hardware tag-delay correction projects a 3D median near 61.7 mm (58.6 mm on the common-mode diagnostic layout). The headline is reported as-measured; these are mechanism-isolation diagnostics, not competing claims.

Two deployment lessons. *Filtering is not calibration* — external position filters cut per-session spread from 67.1 mm to 18.6 mm (§6, Table 8) but leave absolute accuracy essentially unchanged; they remove jitter, not bias. *The dynamic floor is UWB-intrinsic* — RotoArm trajectories show a ∼100 mm median floor (§7, Table 9) that persists across every layout and tag-solver combination and is not removed by substituting Vicon-measured anchor coordinates, pointing to motion-dependent range bias, antenna orientation, and shadowing rather than layout quality.

UWB + IMU fusion (offline replay). With a synthetic IMU derived from the Vicon antenna-centre trajectory, fusion acts primarily as tail control: the best production row reduces track-level P95 from 231.8 mm to 112.1 mm (§11, Table 18), while UWB absolute positions are required to bound inertial drift. This is replay evidence under datasheet-informed sensor models, not hardware-in-the-loop validation.

Limitations (§13). No hardware time synchronisation (dynamic error is best-fit post-processed, not clock-referenced); a single favourable open-lab site; discrete static positions; constrained circular RotoArm motion; simulated rather than embedded IMU.

Next steps (§13). (1) GPIO/event hardware time-sync; (2) per-device antenna-delay bench calibration plus a one-time or windowed tag-side delay correction, as the dominant vertical-error lever; (3) DOP-aware runtime confidence conditioned on the surviving anchor geometry; (4) closed-loop hardware IMU (ICM-45686 class); (5) unconstrained 3D motion and $\geq$1 cluttered environment; (6) independent ultrasound height-gauge verification.

Bottom line (§14). Within this dataset, AutoPos delivers sub-decimetre median static localisation from a self-calibrated anchor array (up to a manually resolved chirality) in a favourable environment. Two concrete obstacles remain on the path to clinical full-body capture: the apparent scale bias (traced to per-device range bias, largely recoverable by solver re-parameterisation plus delay calibration) and the ∼100 mm dynamic floor (addressed by time-sync and hardware IMU). Neither is resolved by this dataset.